MEL G624 (sem -1/2026-27)

Assignment-1

Date of assignment issue: **October 24, 2026**

Date of assignment submission **November 4, 2026**

Maximum Marks: 10

SIMD extensions to CPU architectures as well as ISAs of GPUs use execution stage logic that can perform operations on 32-bit registers that may each contain a single 32-bit-size operand or perform parallel multiple operations on packed multiple smaller bit-size (8/16-bit) operands. The idea of this assignment is to design optimal logic for such an execution stage for a rather limited specification given below:

Specification:

Design execution stage logic which can perform (under 2-bit function control input fn) unsigned addition (fn = 00) / unsigned multiplication (fn = 01) / equality test operation (fn = 10) on two 8-bit inputs A and B by treating each input as a single data item of 8-bit size (if the size control input bit = 0) and treating each input as containing two packed 4-bit sized operands (if the size control input bit = 1) to generate two 8-bit outputs C and D that contain the result of operation as per following format:

For 8-bit addition, D contains the sum bits resulting from the addition after discarding the carry-out bit, and all bits of C are zeros.

For 8-bit multiplication D contains the lower significant bits of the result and C contains the higher significant bits of the result

For 8-bit equality test, lsb of D reflects the result of comparison while other bits of D and all the bits of C are zeros.

For 4-bit addition, D [3 : 0] contain the sum bits of A [3:0] + B [3:0] and D[7:4] contain the sum bits of A [7:4] + B [7:4] and all bits of C are zeros.

For 4-bit multiplication, D contains the result of multiplication of lower nibbles A [3:0] x B [3:0] and C contains the result of multiplication of upper nibbles.

For 4-bit equality test, bit 0 and bit 4 of D reflect the results of equality test of lower and upper nibbles and the remaining bits of D and all the bits of C are zeros.

Explore and design the optimal hardware solution using Verilog modeling and synthesis, and characterize your proposed solutions for speed and gate count.

Prepare a report that explains your explorations and presents the characterization results that can help one decide which solution can be optimal for one's needs. Include your Verilog models in the report.